Vector Fields and Line Integrals

Lecture 01

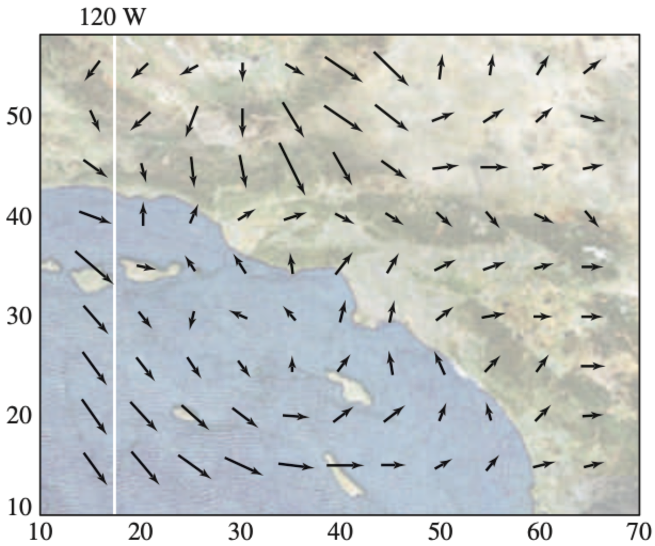
Instructor: Asgher Ali

August 28, 2025

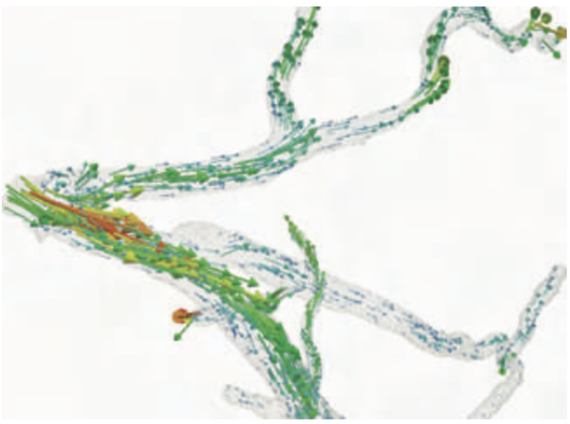
Vector Field

How can we describe a physical object such as the wind, which consists of a large number of molecules moving in a region of space?

What we need is a new type of function called a **vector field**. In this case, a vector field $\mathbf{F}$ assigns to each point $P = (x, y, z)$ a vector $\mathbf{F}(x, y, z)$ that represents the velocity (speed and direction) of the wind at that point.



Another example of vector field is the velocity field driven by the blood flow in an artery.



However, vector fields describe many other types of quantities, such as forces, and electric and magnetic fields.

Mathematically, a **vector field** is a function that assigns a vector to each point in its domain. A vector field on a three-dimensional domain in space will have the formula

$$\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$$

or

$$\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z)\rangle$$

Examples

- A spin field $\mathbf{F} = \dfrac{(-y\mathbf{i} + x\mathbf{j})}{\sqrt{x^2 + y^2}}$
- The radial field $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$
- $\mathbf{F} = \mathbf{i} + x\mathbf{j}$

Hint: You can sketch but that will take quite a while or you can seek help from calcplot3d

Gradient Field

Recall: The gradient vector of a differentiable scalar valued function at a point gives the direction of the greatest increase of the function.

- **Gradient field** is formed by all the gradient vectors of the function.
- Mathematically, the gradient field of a differentiable function $f(x, y, z)$ the field of gradient vectors

$$\vec{\nabla} f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$$

- At each point (x, y, z), the gradient field gives a vector pointing in the direction of greatest increase of f, with magnitude being the value of the directional derivative in that direction.

Example: We need a scalar function, let's say $f(x, y) = x^2y^2$. We can have a gradient field F consisting all the vectors defined on all points (x, y) in the domain of f such that

$$\vec{\nabla} f = 2xy^2\mathbf{i} + 2x^2y\mathbf{j}$$

Motivation

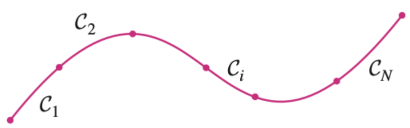
In this section, we define an integral that is similar to a single integral except that instead of integrating over an interval $[a, b]$, we integrate over a curve C. Such integrals are called line integrals, although "curve integrals" would be better terminology. They were invented in the early 19th century to solve problems involving fluid flow, forces, electricity, and magnetism.

Scalar Line Integral

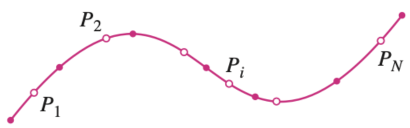
We begin by defining the **scalar line integral**

$$\int_C f(x, y, z)ds$$

of a function $f(x, y, z)$ over a curve C. We will see how integrals of this type represent total mass and charge, and how they can be used to find electric potentials. Let's see the the curve C



Partition of C into N small arcs



Choice of sample points P_i in each arc

Like all integrals, this line integral is defined through a process of subdivision, summation, and passage to the limit. We divide C into N consecutive arcs $C_1, C_2, \cdots, C_N$, choose a sample point P_i in each arc C_i , and form the Riemann sum

$$\sum_{i=1}^{N} f(P_i) \text{ length}(C_i) = \sum_{i=1}^{N} f(P_i)\Delta s_i$$

Mathematically, applying the limit on Δs_i, approaches to zero, we get

$$\int_C f(x,y,z)ds = \lim_{\Delta s_i \to 0} \sum_{i=1}^{N} f(P_i)\Delta s_i$$

In Calculus II, we found that the length of C (provided you have a parametrization) is

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt.$$

A similar type of argument can be used to show that if f is a continuous function, then the limit always exists, and the following formula can be used to evaluate the line integral:

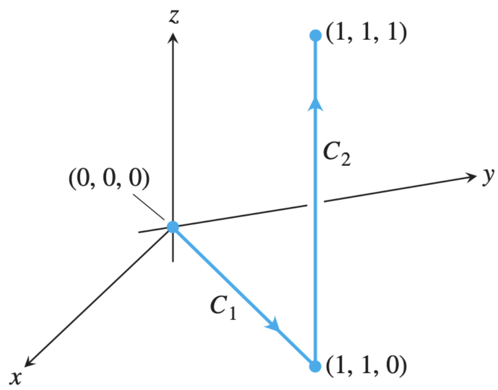
$$\int_C f(x,y,z)\, ds = \int_a^b f(x(t), y(t)), z(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt.$$

Computing a scalar line integral

Let $\mathbf{r(t)}$ be a parameterization of a curve C for $a \leq t \leq b$. If $f(x,y,z)$ and $\mathbf{r}'(t)$ are continuous then

$$\int_C f(x,y,z)ds = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt$$

Example



The figure above show a path from the origin to $(1,1,1)$, the union of line segments C_1 and C_2. Integrate $f(x,y,z) = x - 3y^2 + z$ over $C_1 \cup C_2$.

Step-1: We choose the simplest parametrizations for C_1 and C_2 we can find, calculating the lengths of the velocity vectors as we go along:

$$\begin{aligned} C_1 : \mathbf{r}_1(t) &= \langle 0,0,0\rangle + t\{\langle 1,1,0\rangle - \langle 0,0,0\rangle\} \\ &= t\mathbf{i} + t\mathbf{j}, \quad 0 \le t \le 1 \end{aligned}$$

and

$$\begin{aligned} C_2 : \mathbf{r}_2(t) &= \langle 1,1,0\rangle + t\{\langle 1,1,1\rangle - \langle 1,1,0\rangle\} \\ &= \mathbf{i} + \mathbf{j} + t\mathbf{k}, \quad 0 \le t \le 1 \end{aligned}$$

Step-2: Computing the length of velocity vector

$$\begin{aligned} |\mathbf{v}_1(t)| &= \sqrt{1^2+1^2} = \sqrt{2} \\ |\mathbf{v}_2(t)| &= \sqrt{0^2+0^2+1^2} = 1 \end{aligned}$$

Step-3: With these parametrizations we find that

$$\begin{aligned}
\int_{C_1 \cup C_2} f(x,y,z)ds &= \int_{C_1} f(x,y,z)ds + \int_{C_2} f(x,y,z)ds \\
&= \int_0^1 f(t,t,0)\sqrt{2}dt + \int_0^1 f(1,1,t)1dt \\
&= \int_0^1 (t - 3t^2 + 0)\sqrt{2}dt + \int_0^1 (1 - 3 + t)dt \\
&= -\frac{1}{\sqrt{2}} - \frac{3}{2}
\end{aligned}$$

Cool Animation.

Example

Find the line integral of $f(x,y,z) = 2xy + \sqrt{z}$ over the helix $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$, $0 \leq t \leq \pi$.

Applications: Mass and Moment Calculations

We treat coil springs and wires as masses distributed along smooth curves in space. The distribution is described by a continuous density function $\delta(x, y, z)$ representing mass per unit length.

When a curve C is parametrized by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}, \qquad a \leq t \leq b,$$

then x, y, and z are functions of the parameter t, the density is the function $\delta(x(t), y(t), z(t))$, and the arc length differential is given by

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt.$$

The spring's or wire's mass, center of mass, and moments are then calculated with the formulas in Table 16.1, with the integrations in terms of the parameter t over the interval $[a, b]$. For example, the formula for mass becomes

$$M = \int_a^b \delta(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt.$$

These formulas also apply to thin rods, and their derivations are similar to those in Section 6.6 (Chapter 6).

Mass and moment formulas for coil springs, wires, and thin rods lying along a smooth curve C in space

---

Mass:

$$M = \int_C \delta \, ds, \qquad \delta = \delta(x, y, z) \text{ is the density at } (x, y, z)$$

First moments about the coordinate planes:

$$M_{yz} = \int_C x\delta \, ds, \quad M_{xz} = \int_C y\delta \, ds, \quad M_{xy} = \int_C z\delta \, ds$$

Coordinates of the center of mass:

$$\bar{x} = \frac{M_{yz}}{M}, \qquad \bar{y} = \frac{M_{xz}}{M}, \qquad \bar{z} = \frac{M_{xy}}{M}$$

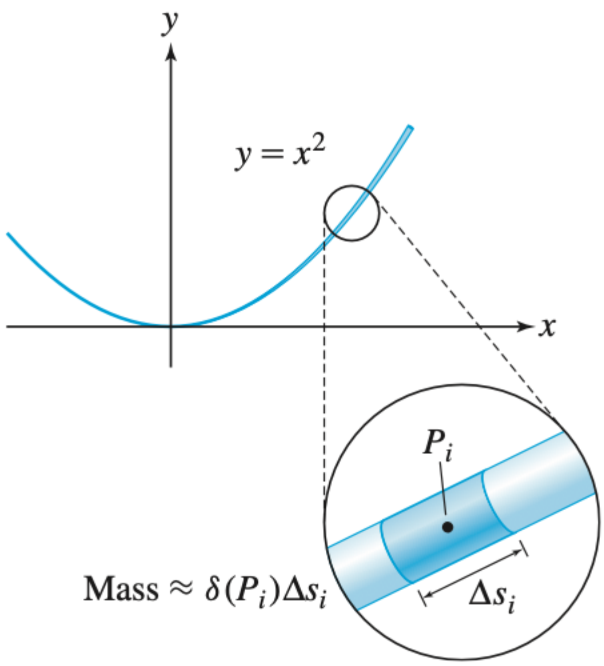
Moments of inertia about axes and other lines:

$$I_x = \int_C (y^2 + z^2)\delta \, ds, \quad I_y = \int_C (x^2 + z^2)\delta \, ds, \quad I_z = \int_C (x^2 + y^2)\delta \, ds$$

$$I_L = \int_C r^2 \delta \, ds, \qquad r(x, y, z) = \text{distance from } (x, y, z) \text{ to line } L$$

Application: Mass of a Wire

Find the total mass of a wire in the shape of the parabola $y = x^2$ for $1 \leq x \leq 4$ (in centimeters) with mass density given by $\delta(x, y) = \frac{y}{x}$ g/cm.



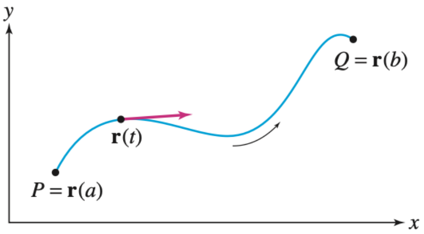
Vector Line Integrals

When you carry a backpack up a mountain, you do work against the earth's gravitational field. The work, or energy expended, is one example of a quantity represented by a **vector line integral**.

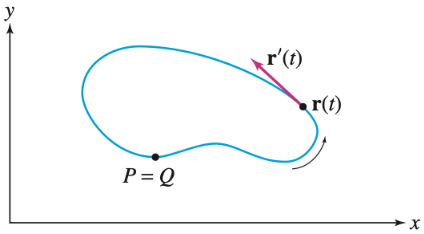
An important difference between vector and scalar line integrals is that vector line integrals depend on the direction along the curve.

This is reasonable if you think of the vector line integral as work, because the work performed going down the mountain is the negative of the work performed going up.

A specified direction along a curve C is called an **orientation**. We refer to this direction as the positive direction along C, the opposite direction is the negative direction, and C itself is called an **oriented curve**.



(A) Oriented path from P to Q

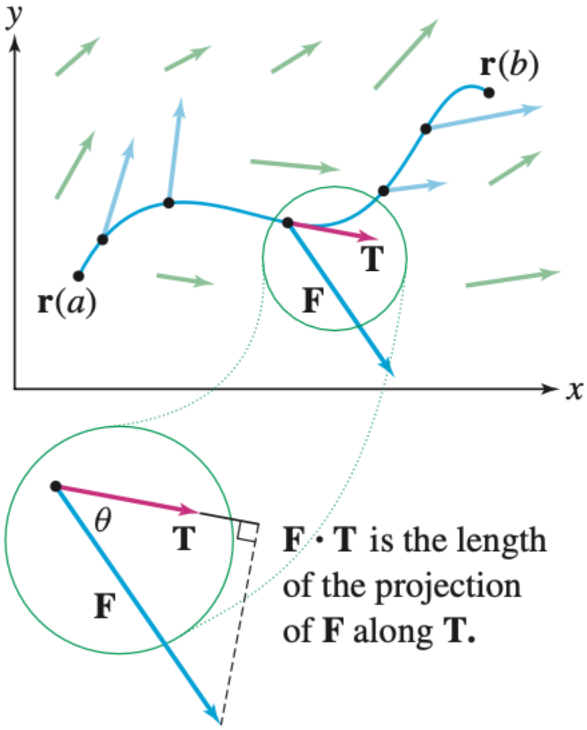


(B) A closed oriented path

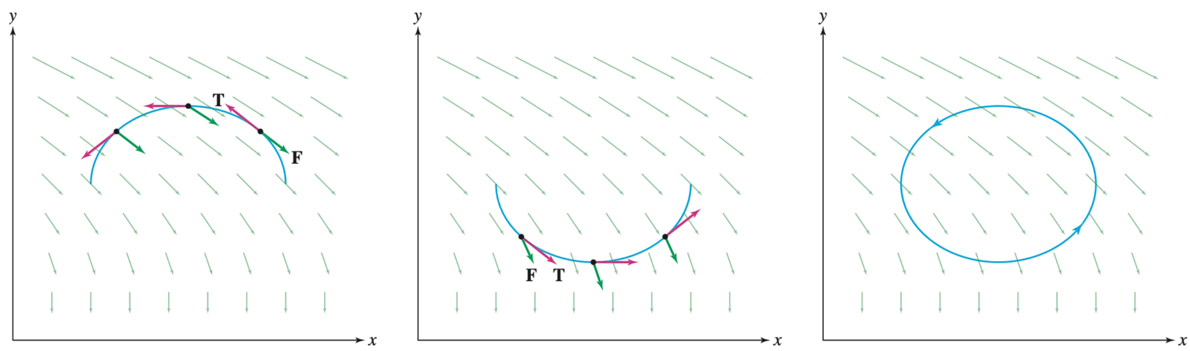
Definition

The line integral of a vector field $\mathbf{F}$ along an oriented curve C is the integral of the tangential component of F:

$$\int_C f(x,y,z)ds = \int_C (\mathbf{F}\cdot\mathbf{T})ds = \int_C \mathbf{F}\cdot d\mathbf{r}$$



The graphical insight we get



(A) Most of the dot products $\mathbf{F}\cdot\mathbf{T}$ are negative because the angles between the vectors are obtuse. Therefore, the line integral is negative.

(B) Most of the dot products $\mathbf{F}\cdot\mathbf{T}$ are positive because the angles between the vectors are acute. Therefore, the line integral is positive.

(C) Total line integral is negative.

We use parametrizations to evaluate vector line integrals, but there is one important difference from the scalar case: the **parametrization $\mathbf{r}(t)$ must be positively oriented; that is, $\mathbf{r}(t)$ must trace C in the positive direction.**
We also assume that $\mathbf{r}(t)$ is ***regular***; that is,

$$\mathbf{r}'(t) \neq 0 \quad \text{for } a \leq t \leq b.$$

Then $\mathbf{r}'(t)$ is a nonzero tangent vector pointing in the positive direction, and

$$\mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}.$$

In terms of the arc length differential

$$ds = |\mathbf{r}'(t)|\, dt,$$

we have

$$(\mathbf{F} \cdot \mathbf{T})\, ds = \mathbf{F}(\mathbf{r}(t)) \cdot \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}\, |\mathbf{r}'(t)|\, dt = \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt.$$

Computing a Vector Line Integral

If $\mathbf{r}(t)$ is a regular parametrization of an oriented curve C for $a \leq t \leq b$, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Examples

1. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle z, y^2, x \rangle$ and C is parametrized (in the positive direction) by $\mathbf{r}(t) = \langle t+1, e^t, t^2 \rangle$ for $0 \leq t \leq 2$.

2. The ellipse in the figure above with counterclockwise orientation is parametrized by $\mathbf{r}(\theta) = \langle 5 + 4\cos(\theta), 3 + 2\sin(\theta) \rangle$ for $0 \leq \theta \leq 2\pi$. Calculate

$$\int_C 2y dx - 3dy$$

3. Compute

$$\int_C \mathbf{F} \cdot d\mathbf{r},$$

where

$$\mathbf{F}(x, y, z) = \langle e^z,\ e^y,\ x + y \rangle,$$

and C is the triangle joining the points

$$A = (1, 0, 0), \quad B = (0, 1, 0), \quad C = (0, 0, 1),$$

oriented counterclockwise when viewed from above.

Note: The triangle is piecewise smooth; it is the union of its three smooth edges.

Properties of Vector Line Integrals

Let C be a smooth oriented curve, and let $\mathbf{F}$ and $\mathbf{G}$ be vector fields.

(i) **Linearity:**

$$\int_C (\mathbf{F} + \mathbf{G}) \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot d\mathbf{r} + \int_C \mathbf{G} \cdot d\mathbf{r},$$

$$\int_C k\mathbf{F} \cdot d\mathbf{r} = k \int_C \mathbf{F} \cdot d\mathbf{r}, \quad (k \in \mathbb{R}).$$

(ii) **Reversing Orientation:**

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}.$$

(iii) **Additivity:** If C is the union of n smooth curves $C_1, C_2, \ldots, C_n$, then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} + \cdots + \int_{C_n} \mathbf{F} \cdot d\mathbf{r}.$$

Application of the Vector Line Integral

Recall that in physics, "work" refers to the energy expended when a force is applied to an object as it moves along a path. By definition, the work W performed along the straight segment from P to Q

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}$$

Calculating Work

Calculate the work performed against $\mathbf{F}$ in moving a particle from $P = (0,0,0)$ to $Q = (4,8,1)$ along the path $\mathbf{r}(t) = \langle t^2, t^3, t \rangle$ (in meters), for $1 \leq t \leq 2$ in the presence of the a force field

$$\mathbf{F} = \langle x^2, -z, -yz^{-1} \rangle$$

in newtons.

Hint: $W = -\int_C \mathbf{F} \cdot d\mathbf{r}$

Example

Calculate the work done by a field $\mathbf{F} = \langle x + y, x - y \rangle$ when an object moves from $(0,0)$ to $(1,1)$ along each of the paths $y = x^2$ and $x = y^2$.